Inorganic Chemistry

Notes

Classification of Elements

- Periodic Table or Periodic table of elements is a tabular arrangement of the chemical elements, arranged by atomic number, electron configuration and recurring chemical properties, whose structure shows periodic trends.

Need for classification of Elements :

- Before the beginning of the 18th century, only a few elements were known, so it was quite easy to study and remember the properties of those elements and their compounds individually.
- Later on, with the increasing number of elements, it became more and more difficult to study their properties individually, therefore the need for their classification was felt.

Development of Periodic Table :

- Several major attempts made for classification of elements. Some of them are as follows -

(i) Prout Hypothesis (1815) –

- William Prout observed that the atomic weights that had been measured for the elements known at that time appeared to be whole multiples of the atomic weight of hydrogen.
- He hypothesized that the hydrogen atom was the only truly fundamental object, which he called protyle and that the atoms of other elements were actually groupings of various numbers of hydrogen atoms.
- This postulation was discarded when the atomic weight of elements not found in the whole number.

(ii) Dobereiner's Triads (1829) –

- According to this law, when chemically analogous elements are arranged in order of increasing atomic mass, they formed well-marked group of three called 'Triads', in which the atomic mass of middle element was nearly equal to the arithmetic mean of the other.
- Examples of triads of elements are as follows –

	Element	Atomic Mass	Mean of 1 & 3
1.	Lithium	7	
2.	Sodium	23	$\frac{7+39}{2} = 23$
3.	Potassium	39	
1.	Calcium	40	
2.	Strontium	88	$\frac{40+137}{2} = 88.5$
3.	Barium	137	

- This classification did not receive wide acceptance since only a few elements could be arranged into triads.

(iii) Newlands' Law of Octaves (1865) –

- If the chemical elements are arranged according to increasing atomic weight, those with similar physical and chemical properties occur after each interval of seven elements.
- Arrangement of some elements with their atomic weights according to the Law of Octaves **:**

Li	Be	B	C	N	O	F
(7)	(9)	(11)	(12)	(14)	(16)	(19)
Na	Mg	Al	Si	P	S	Cl
(23)	(24)	(27)	(28)	(31)	(32)	(35.5)

- Newlands contributed the word 'periodic' in chemistry.

(iv) Lothar Meyer's Atomic Volume Curve (1869) –

- Lothar Meyer, a German chemist plotted a graph between atomic weight and atomic volume (i.e. atomic weight in solid state / density), he found that elements with similar properties occupied the similar position on the graph.

(v) Mendeleev's Periodic Law –

- "The physical and chemical properties of the elements are a periodic function of their atomic mass".

Salient Features of Mendeleev's Periodic Table :

(i) Mendeleev arranged the elements in the increasing order of their atomic mass.

(ii) The elements are arranged in horizontal rows called periods. Those are six in numbers (1 to 6 - Arabic numerals). Period 4th, 5th & 6th have two series of elements.

(iii) The vertical columns are called groups. There are eight groups numbered from I to VIII (Roman Numerals).

(iv) Groups I to VII are further divided into A & B subgroups. However, group VIII contains three elements in each of the three periods.

(v) All the elements present in a particular group are chemically similar in nature. They also show a regular gradation in their physical and chemical properties form top to bottom.

(vi) Mendeleev's classification included all the 63 elements known at that time.

(vii) He left blank spaces for elements yet to be discovered, which were filled with the discovery of germanium, scandium and gallium.

(vi) Modern Periodic Table –

- Keeping in view the drawbacks of Mendeleev's periodic table such as anomalous pair - placing of heavier element before lighter one and there is no place for different isotopes of an element in the periodic table, it was felt that the arrangement of elements in the periodic table should be based on some other property which is more fundamental than atomic mass.
- **Moseley** (1913) arranged the elements in a tabular form based on their atomic numbers.
- It states that the chemical and physical properties of elements are a periodic function of their atomic number i.e if elements are arranged in the order of their increasing atomic number, the elements with similar properties are repeated after certain regular intervals.

(vii) Long Form of Periodic Table –

- The accepted modern periodic table is the Long form of Periodic Table.
- The arrangement of elements in this table is also in keeping with their electronic structure (configuration). Columns represent the Groups or family and rows represent the Periods.

Groups :

- It contains 18 vertical columns. The groups have been numbered from 1 to 18 (in Arabic numerals).
- All elements present in a group generally have similar electronic configuration in their valence shell and have a same number of valence electrons.

Periods :

- There are seven horizontal rows in the modern periodic table.
- The elements in a period have consecutive atomic numbers.
- The periods have been numbered form 1 to 7 (Arabic numerals).
- In each period a new shell starts filling up. The period number is also the number of shells which starts filling up as we move from left to right across the period.

(a) The first period is the shortest period of all. It contains only two elements – H and He.

(b) The second and third periods are called short-periods each with 8 elements.

(c) The 4th & 5th periods are called long periods each with 18 elements.

(d) The 6th & 7th periods are called very long periods containing 32 elements each.

Types of Elements :

1. Main Group Elements or Normal Elements –

- Main Group elements have been placed in groups 1 and 2 on the left side and in groups 13 to 17 on the right side of the periodic table.
- Their outermost shell is incomplete (their outermost shell has less than eight electrons).

2. Noble Gases –

- Group 18 contains noble gases. Their outermost shell contains 8 electrons except Helium (only 2 electron in the outermost shell).
- These are inert gases (do not react) and with zero valency.

3. Transition Elements (d-block elements) –

- Groups 3 to 12 contain transition elements.
- Their two outermost shells are incomplete.
- All these elements are metals with high melting and boiling points, they are a good conductor of heat and electricity, magnetic and exhibit variable valancies.

4. Inner Transition Elements (f-block elements) –

- These have been shown separately below the main periodic table.
- Their three outermost shells are incomplete.
- There are two series of 15 elements.

a . Lanthanoids (4f)

- This series consists of 15 elements from Atomic No. 57 to 71 (La to Lu).
- These are all placed in group 3, period 6.
- These elements, along with the chemically similar elements scandium and yttrium, are often collectively known as the 'rare earth elements'.

b . Actinoids (5f)

- It consists of elements from atomic number 89 to 103 (Ac to Lr).
- These are all placed in group 3, Period 7.
- All actinoids are radioactive and release energy upon radioactive decay.
- The element Lanthanum (57) and Actinium (89) are the eponyms of Lanthanoid and Actinoid series respectively.
- Even though lanthanoid means like lanthanum and actinoid means like actinium and as such they should not be included in these groups but IUPAC acknowledges their inclusion based on common usage.
- Sometimes these groups are considered to have 14 elements each (without lanthanum and actinium).

Periodicity in Properties of Elements

S. N.	Property	In Period (Form left to Right)	In Groups (From top to bottom)
1.	Atomic Size	Decreases	Increases
2.	Atomic Radius	Decreases	Increases
3.	Ionic Radius	Decreases	Increases
4.	Ionic Potential	Increases	Decreases
5.	Electronegativity	Increases	Decreases
6.	Electron Affinity	Increases	Decreases
7.	Melting point	Increases	Decreases
8.	Nature of Hydrides	Becomes acidic from alkaline	Increasing in acidic nature
9.	Nature of Oxides	Alkinity decreases and acidity increases	Alkalinity Increases and acidity decreases
10.	Metallic characteristics	Decreases	Increases
11.	Oxidising nature	Increases	Decreases

Name of Elements from Atomic Number 101 to 118

Atomic Number	IUPAC Name	IUPAC Symbol
101	Mendelevium	Md
102	Nobelium	No
103	Lawrencium	Lr
104	Rutherfordium	Rf
105	Dubnium	Db
106	Seaborgium	Sg
107	Bohrium	Bh
108	Hassium	Hs
109	Meitnerium	Mt
110	Darmstadtium	Ds
111	Roentgenium	Rg
112	Copernicium	Cn
113	Nihonium	Nh
114	Flerovium	Fl
115	Moscovium	Mc
116	Livermorium	Lv
117	Tennessine	Ts
118	Oganesson	Og

Question Bank

1. UNESCO inaugurated the celebration of 2019 as the International Year of the Periodic Table of Chemical Elements to celebrate its completion of how many years?

(a) 100 (b) 150

(c) 75 (d) 50

(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

UNESCO inaugurated the celebration of 2019 as the International Year of the Periodic Table of Chemical Elements to celebrate its completion of 150 years. The year 2019 marked the 150th anniversary of the creation of periodic table by Russian scientist Dmitry Ivanovich Mendeleev in 1869.

2. Assertion (A): In the periodic table of chemical elements, electron affinity is always found to increase from top to bottom in a group.

Reason (R) : In a group, the atomic radii generally increase from top to bottom.

(a) Both (A) and (R) are individually true, and (R) is the correct explanation of (A).

(b) Both (A) and (R) are individually true, but R is not the correct explanation of (A).

(c) (A) is true, but (R) is false.

(d) (A) is false, but (R) is true.

I.A.S. (Pre) 2003

Ans. (d)

In the periodic table of chemical elements, the electron affinity decreases down the group but increases up for the group and from left to right across periods of a periodic table because the electron added to energy levels become closer to the nucleus, thus a stronger attraction between the nucleus and its electrons. An atom gets larger as the number of electronic shells increases. Therefore, the radius of atoms increases as you go down a certain group in the periodic table of elements. However, the size of an atom will decrease as you move from left to right of a certain period. Thus, Assertion (A) is false, while Reason (R) is true.

3. Consider the following statements with reference to the periodic table of chemical elements :

1. Ionization potential gradually decreases along a period.

2. In a group of elements, electron affinity decreases as the atomic weight increases.

3. **In a given period, electronegativity decreases as the atomic number increases.**

Which of these statement (s) is/are correct ?

(a) 1 only (b) 2 only

(c) 1 and 3 (d) 2 and 3

I.A.S. (Pre) 2001

Ans. (b)

The ionization energies are dependent upon the atomic radius. Since going from left to right across a period in the periodic table, the atomic radius decreases and the ionization potential increases while ionization potential decreases as one moves down a given group.
In a group of elements, from top to bottom the electron affinity decreases as the atomic weight increases. As one moves trend-wise from left to right across a period in the periodic table, the electronegativity increases as the atomic number increases, due to the stronger attraction that the atoms obtain as the nuclear charge increases.

4. **What is the usual property of oxides of Group 3 and 4?**

(a) Basic and Acidic (b) Basic

(c) Acidic (d) Neutral

43rd B.P.S.C. (Pre) 1999

Ans. (a)

The usual property of oxides of Group 3 and 4 of the periodic table are basic and acidic. These oxides are amphoteric (able to react both as a base and as an acid).

5. **Correctly match the following :**

Elements	**Valency**
A. Silicon	**(i) 1**
B. Fluorine	**(ii) 2**
C. Aluminium	**(iii) 3**
D. Sulphur	**(iv) 4**

Code :

	A	B	C	D
(a)	(iv)	(iii)	(ii)	(i)
(b)	(iv)	(i)	(iii)	(ii)
(c)	(iii)	(iv)	(i)	(ii)
(d)	(iii)	(iv)	(ii)	(i)

Chhattisgarh P.C.S. (Pre) 2022

Ans. (b)

The correctly matched lists are as follows :

Elements		**Valency**
Silicon	–	4
Fluorine	–	1
Aluminium	–	3
Sulphur	–	2

Hence, option (b) is the correct answer.

6. **The element found maximum in the soil layer is –**

(a) Oxygen (b) Nitrogen

(c) Manganese (d) Silicon

42nd B.P.S.C. (Pre) 1997

Ans. (a)

The element found the maximum in the soil layer or Earth's crust as mass percentage (about) is Oxygen 46.60% followed by Silicon 27.72%, Aluminium 8.13%, Iron 5.00% , Calcium 3.65% and Carbon 0.6%.

7. **Which one of the following is present in the largest amount in terms of percent by mass in the Earth's crust?**

(a) Silicon (b) Oxygen

(c) Carbon (d) Calcium

I.A.S. (Pre) 1997

Ans. (b)

See the explanation of above question.

8. **Which is the most abundant element after Oxygen?**

(a) Silicon (b) Carbon

(c) Sodium (d) Chlorine

M.P.P.C.S. (Pre) 2005

Ans. (a)

The most abundant element on Earth's surface after Oxygen is Silicon. It was discovered by J.J. Berzelius in 1824. The word 'Silicon' was taken from the Latin word silex. Silicon chips are used as a semiconductor in computers.

9. **Which of the following is the most common element in the Universe?**

(a) Hydrogen (b) Oxygen

(c) Nitrogen (d) Carbon

U.P.P.C.S. (Pre) 2007

Ans. (a)

The most abundant element in the Universe is Hydrogen, which makes up about 3/4 of all matter. Helium makes up most of the remaining 1/4. Thus, it is clear that hydrogen is the most common and abundant element in the Universe. While the most abundant element in the Earth's crust is Oxygen making up 46.6% of Earth's crust.

10. **Approximately how many different chemical elements exist on the Earth?**

(a) 300 (b) 250

(c) 200 (d) 100

R.A.S./R.T.S. (Pre) 2003

Ans. (*)

A chemical element is a substance which consists of atoms having the same number of protons in their atomic nuclei. There are 118 elements that have been identified till date. On 28 November, 2016, the International Union of Pure and